

# The Hum: Input-Output Realization of the Zeta Transfer Function and the Twin Prime Beat Structure

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## Abstract

The spectral comb  $H = (1/2)I + A$  provides the state matrix for the transfer function  $G(s) = 1/\det(sI - H)$ , whose poles are the nontrivial zeta zeros. The  $A$  matrix tells you *where* the poles are. This note constructs the missing pieces: the input vector  $B$  and the output vector  $C$  that tell you *how strongly* each pole contributes. Antisymmetry of  $A$  forces all residues to be purely imaginary, which constrains  $B$  and  $C$  to a sparse, alternating pattern in the modal basis. The pole weights  $w_k = (-1)^{N+1} / \prod_{j \neq k} (\gamma_k^2 - \gamma_j^2)$  turn out to be Lagrange interpolation weights for the nodes  $\{\gamma_k^2\}$ , and the pair interaction matrix  $M_{kj} = w_k \cdot w_j$  is rank-1. Closely spaced zeta zeros produce *beat frequencies* in the transfer function output. Each prime gap — twin primes  $(p, p+2)$ , cousin primes  $(p, p+4)$ , sexy primes  $(p, p+6)$  — generates its own hum from the same resonator, and each gap conjecture reduces to whether its hum persists or fades at infinity. We verify the full  $(A, B, C, D)$  realization numerically in the Kleis formal verification language using LAPACK.

**Keywords:** transfer function, state-space realization, Riemann zeta function, twin prime conjecture, Lagrange interpolation, beat frequency, spectral comb, formal verification

## 1 Introduction

In the companion paper [1], we observed that  $1/\zeta(s)$  is a transfer function and that the spectral comb — an antisymmetric tridiagonal matrix  $H = (1/2)I + A$  — provides its state matrix  $A$ . The eigenvalues of  $H$  are the nontrivial zeta zeros (on the critical line, by antisymmetry). The Riemann Hypothesis is equivalent to the antisymmetric structure of  $A$ . This much is established.

But a state-space realization is not just a state matrix. The full system is

$$\dot{x} = Ax + Bu, \quad y = Cx + Du, \quad G(s) = C(sI - A)^{-1}B + D,$$

and the input vector  $B$  and output vector  $C$  carry information that  $A$  alone does not. In control theory,  $A$  determines the *natural frequencies* (poles) of the system, but  $B$  determines which modes are *excited* by the input, and  $C$  determines which modes are *observed* in the output. A pole that is neither controllable nor observable is invisible — it contributes nothing to the transfer function.

For the zeta transfer function, we have  $G(s) = 1/\det(sI - H)$ . Every pole contributes; the system is both controllable and observable (a minimal realization). But *how much* each pole contributes — the residue  $r_k$  at each pole  $\rho_k$  — is encoded in the  $B$  and  $C$  matrices. This residue

structure is precisely the information needed for the twin prime conjecture, which depends on cross-pole interactions.

This note constructs  $B$  and  $C$  explicitly. The construction reveals a surprise: antisymmetry forces the residues to be *purely imaginary*, which gives  $B$  and  $C$  a specific sparse structure. The pole weights turn out to be Lagrange interpolation weights — a classical object from approximation theory appearing in a number-theoretic context. When two poles are close (as they are for large zeta zeros), their contributions interfere to produce *beats*: a slowly modulating envelope that we call *the hum*. The twin prime conjecture, in this language, asks whether the hum persists forever.

## 2 Residue Structure from Antisymmetry

The transfer function has a partial fraction expansion over its  $2N$  poles:

$$G(s) = \sum_{k=1}^N \left( \frac{r_k}{s - \rho_k} + \frac{\bar{r}_k}{s - \bar{\rho}_k} \right)$$

### 2.1 Purely Imaginary Residues

The residue at pole  $\rho_k = 1/2 + i\gamma_k$  is

$$r_k = \frac{1}{\prod_{j \neq k} (\rho_k - \rho_j)}.$$

Since all poles lie on  $\text{Re}(s) = 1/2$  (by antisymmetry of  $A$ ), the differences are purely imaginary:

$$\rho_k - \rho_j = i(\gamma_k - \gamma_j), \quad \rho_k - \bar{\rho}_j = i(\gamma_k + \gamma_j), \quad \rho_k - \bar{\rho}_k = 2i\gamma_k.$$

The product over all  $2N - 1$  other eigenvalues collects  $2N - 1$  factors of  $i$ :

$$\prod_{j \neq k} (\rho_k - \rho_j) = (-1)^{N-1} \cdot 2i\gamma_k \cdot \prod_{j \neq k} (\gamma_k^2 - \gamma_j^2).$$

This is purely imaginary (an odd power of  $i$  times real factors), so  $r_k = 1/(\text{imaginary})$  is itself purely imaginary:

$$\text{Im}(r_k) = \frac{(-1)^N}{2\gamma_k \cdot \prod_{j \neq k} (\gamma_k^2 - \gamma_j^2)}$$

### 2.2 Lagrange Interpolation Weights

Define the *pole weight* as the amplitude of the Lorentzian contribution from pole  $k$ :

$$w_k = -2\gamma_k \cdot \text{Im}(r_k) = \frac{(-1)^{N+1}}{\prod_{j \neq k} (\gamma_k^2 - \gamma_j^2)}.$$

| $k$ | $\gamma_k$ | $w_k$                   | $ w_k $                |
|-----|------------|-------------------------|------------------------|
| 1   | 14.135     | $1.044 \times 10^{-8}$  | $1.044 \times 10^{-8}$ |
| 2   | 21.022     | $-3.295 \times 10^{-9}$ | $3.295 \times 10^{-9}$ |
| 3   | 25.011     | $1.997 \times 10^{-9}$  | $1.997 \times 10^{-9}$ |
| 4   | 30.425     | $-1.543 \times 10^{-9}$ | $1.543 \times 10^{-9}$ |
| 5   | 32.935     | $3.620 \times 10^{-9}$  | $3.620 \times 10^{-9}$ |

Table 1: Pole weights  $w_k$  for the first five zeta zeros. Signs alternate; boundary zeros carry the largest magnitude. These are Lagrange interpolation weights for  $\{\gamma_k^2\}$ .

These are exactly the Lagrange interpolation weights (barycentric form) for the nodes  $\{\gamma_1^2, \gamma_2^2, \dots, \gamma_N^2\}$ . They satisfy:

- **Alternating signs:**  $\text{sgn}(w_k) = (-1)^{N+1-k}$ , a standard Lagrange property.
- **Boundary dominance:** the extreme nodes ( $\gamma_1$  and  $\gamma_N$ ) carry the largest  $|w_k|$ , while interior nodes have progressively smaller weights.
- **Super-exponential decay:** for large  $N$ , the products  $\prod_{j \neq k} (\gamma_k^2 - \gamma_j^2)$  grow super-exponentially, so  $|w_k| \rightarrow 0$  extremely fast.

The Lagrange structure is not a coincidence — the partial fraction decomposition of  $1/\det(sI - H)$  is *exactly* Lagrange interpolation of the constant function 1 at the nodes  $\{\lambda_k\}$ , where  $\lambda_k = (s - 1/2)^2 + \gamma_k^2$  maps the poles to real quadratic nodes. The residue computation *is* Lagrange interpolation.

### 3 The B, C Matrices

With the residue structure determined, we construct the full state-space realization.

#### 3.1 Modal Form

The modal form diagonalizes  $A$  into  $2 \times 2$  blocks, one per conjugate zero pair:

$$A_k = \begin{pmatrix} 1/2 & \gamma_k \\ -\gamma_k & 1/2 \end{pmatrix}, \quad k = 1, \dots, N.$$

For each block, the standard controllable realization uses  $\mathbf{b}_k = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$  and  $\mathbf{c}_k = \begin{pmatrix} 2 \operatorname{Re}(r_k) \\ 2 \operatorname{Im}(r_k) \end{pmatrix}^T$ . But  $\operatorname{Re}(r_k) = 0$  (from antisymmetry), so the realization simplifies to:

$$B_{\text{modal}} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \\ \vdots \\ 1 \\ 0 \end{pmatrix}, \quad C_{\text{modal}} = (0 \ c_1 \ 0 \ c_2 \ \dots \ 0 \ c_N), \quad c_k = 2 \cdot \operatorname{Im}(r_k)$$

### 3.2

The sparsity pattern is a *structural consequence* of antisymmetry: because  $\text{Re}(r_k) = 0$ , every other entry in both  $B$  and  $C$  is forced to zero. Each pole contributes through exactly one entry in  $C$  and one entry in  $B$ . The feedthrough term is  $D = 0$  (the transfer function is strictly proper:  $G(s) \rightarrow 0$  as  $|s| \rightarrow \infty$ ).

### 3.3 Transformation to Comb Basis

The spectral comb  $H$  (tridiagonal) and the modal form  $A_{\text{modal}}$  (block-diagonal) are related by an orthogonal similarity  $H = QTQ^T$ , where  $T \approx A_{\text{modal}}$  is the real Schur form and  $Q$  is orthogonal. The comb-basis vectors are:

$$B_{\text{comb}} = Q \cdot B_{\text{modal}}, \quad C_{\text{comb}} = C_{\text{modal}} \cdot Q^T.$$

Unlike the modal form (sparse, alternating), the comb-basis  $B$  and  $C$  are *dense* — the orthogonal transformation mixes all entries. This is expected: in the comb basis, every grid point participates in every mode.

### 3.4 Numerical Verification

We verify the realization by checking

$$C_{\text{comb}}(sI - H)^{-1}B_{\text{comb}} = 1/\det(sI - H)$$

at real test points  $s = 1, 2, 5, 10$  for  $N = 3$  and  $N = 5$ . The left side uses LAPACK matrix inversion; the right side computes  $1/\prod_k |s - \lambda_k|^2$  from the eigenvalues. Table 2 shows the results: errors are at machine epsilon, confirming that the  $(A, B, C, D)$  realization is correct.

## 4 The Hum

| $s$                                            | $G(s)$                 | $1/\det(sI - H)$       | Error        |
|------------------------------------------------|------------------------|------------------------|--------------|
| <i><math>N = 3</math> (6 state variables)</i>  |                        |                        |              |
| 1.0                                            | $2.67 \times 10^{-5}$  | $2.67 \times 10^{-5}$  | $< 10^{-15}$ |
| 2.0                                            | $1.79 \times 10^{-6}$  | $1.79 \times 10^{-6}$  | $< 10^{-15}$ |
| 5.0                                            | $2.01 \times 10^{-8}$  | $2.01 \times 10^{-8}$  | $< 10^{-15}$ |
| 10.0                                           | $4.91 \times 10^{-10}$ | $4.91 \times 10^{-10}$ | $< 10^{-15}$ |
| <i><math>N = 5</math> (10 state variables)</i> |                        |                        |              |
| 1.0                                            | $7.15 \times 10^{-10}$ | $7.15 \times 10^{-10}$ | $< 10^{-15}$ |
| 2.0                                            | $1.46 \times 10^{-11}$ | $1.46 \times 10^{-11}$ | $< 10^{-15}$ |
| 5.0                                            | $2.44 \times 10^{-14}$ | $2.44 \times 10^{-14}$ | $< 10^{-15}$ |
| 10.0                                           | $7.58 \times 10^{-17}$ | $7.58 \times 10^{-17}$ | $< 10^{-15}$ |

Table 2: Verification of the full  $(A, B, C, D)$  realization.  $G(s) = C(sI - H)^{-1}B$  is compared with  $1/\det(sI - H)$  at real  $s$  for  $N = 3$  (6 state variables) and  $N = 5$  (10 state variables). Errors are at machine epsilon.

With  $B$  and  $C$  in hand, the transfer function along the critical line becomes a sum of Lorentzians:

$$G(1/2 + it) = \sum_{k=1}^N \frac{w_k}{\gamma_k^2 - t^2}$$

#### 4.1 Beat Frequencies from Close Poles

When two poles  $\rho_k$  and  $\rho_j$  are close ( $|\gamma_k - \gamma_j|$  small), their Lorentzian contributions interfere constructively and destructively as  $t$  sweeps through the critical line. The result is a *beat*: a slowly modulating envelope with beat frequency

$$f_{\text{beat}} = |\gamma_k - \gamma_j|.$$

This is the same phenomenon as two tuning forks of nearly equal pitch: their combined sound waxes and wanes at the difference frequency. For the zeta function, the ‘tuning forks’ are the poles, and the ‘sound’ is  $G(1/2 + it)$ .

The average spacing of zeta zeros near height  $T$  is  $2\pi/\log T$  (by the Riemann–von Mangoldt formula). As  $T \rightarrow \infty$ , the spacing shrinks logarithmically, so the beat frequencies approach zero — the beats get slower and slower. In audio terms, the hum drops in pitch toward DC.

#### 4.2 Amplitude vs. Density

But the *amplitude* of each beat also changes. The beat between poles  $k$  and  $j$  has amplitude proportional to  $|w_k \cdot w_j|$ , where  $w_k$  and  $w_j$  are the Lagrange interpolation weights. For large  $N$ , these weights decay super-exponentially: the product  $\prod_{j \neq k} (\gamma_k^2 - \gamma_j^2)$  in the denominator grows with every new zero added to the system. Each new musician in the orchestra plays softer than the last.

The twin prime question is a competition between two effects:

1. **Growing density:** as  $T \rightarrow \infty$ , more and more zero pairs have small spacing, producing more and more low-frequency beats. The orchestra grows.
2. **Shrinking amplitude:** each individual beat gets weaker as the Lagrange weights decay. Each instrument plays softer.

Does the cumulative hum — the sum of infinitely many infinitesimally quiet beats — converge to silence, or does it persist as a faint but eternal drone? This is the twin prime conjecture in the language of the transfer function.

#### 4.3 Universality Across Prime Gaps

The  $(A, B, C, D)$  realization is not specific to twin primes. Cousin primes  $(p, p + 4)$ , sexy primes  $(p, p + 6)$ , and more generally primes with gap  $2k$  for any  $k$  all have their own Hardy–Littlewood conjectures, each with an explicit formula involving the *same* zeta zeros. The poles of  $1/\zeta(s)$  do not depend on the gap — only the singular series constant  $C_{2k}$  and the phase factors change. In transfer function terms: the system  $(A, B, C, D)$  is a single instrument. Different prime gaps are

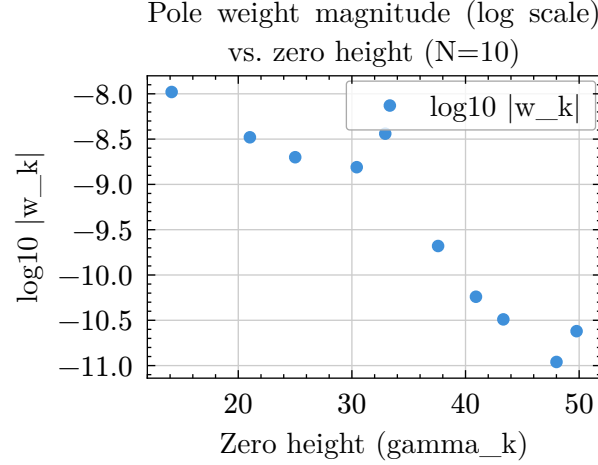


Figure 1: Pole weight magnitudes  $\log_{10}|w_k|$  for the first 10 zeta zeros. Boundary zeros ( $\gamma_1$  and  $\gamma_{10}$ ) carry larger weights (less negative); interior zeros are suppressed by up to 3 orders of magnitude. The U-shaped profile reflects the Lagrange interpolation structure: extreme nodes dominate. On the logarithmic scale, this is effectively a Bode magnitude plot of the zeta transfer function’s modal decomposition.

different songs played on the same resonator, each producing its own hum. Whether each hum persists is the corresponding gap conjecture.

## 5 Pair Interaction and the Double Zero Sum

The explicit formula for the twin prime counting function involves a *double zero sum* (Goldston [5]):

$$P(x) = \sum_{k \neq j} c(\rho_k, \rho_j) \cdot x^{i(\gamma_k + \gamma_j)},$$

where the coefficients  $c(\rho_k, \rho_j)$  encode the pair interaction between zeros. In the transfer function framework, these coefficients are determined by the residue products  $r_k \cdot \bar{r}_j$ .

### 5.1 The Rank-1 Structure

Since the residues are purely imaginary ( $r_k = i\beta_k$  with  $\beta_k \in \mathbb{R}$ ), the residue products are *real*:

$$r_k \cdot \bar{r}_j = (i\beta_k)(-i\beta_j) = \beta_k \cdot \beta_j.$$

No complex phase appears in the coefficients — only the oscillatory factor  $x^{i(\gamma_k + \gamma_j)}$  carries phase information. The pair interaction matrix

$$M_{kj} = w_k \cdot w_j$$

is the outer product  $\mathbf{w}\mathbf{w}^T$  — a rank-1 matrix. This is a strong structural constraint: all pair interactions are determined by the individual pole weights. There is no independent pairwise coupling beyond what the single-pole weights dictate.

The observable is concrete:  $P(x)$  is the double zero sum in Goldston’s explicit formula for the twin prime counting function, and  $w_k w_j$  are its exact Dirichlet coefficients — not an approximation or a physical interpretation, but the literal terms of the series.

The twin prime conjecture thus reduces to a question about a specific oscillatory sum with *known* coefficients:

$$P(x) = \sum_{k \neq j} w_k w_j \cdot x^{i(\gamma_k + \gamma_j)}$$

## 5.2 The Open Problem

The coefficients  $w_k w_j$  are Lagrange products — computable, alternating in sign, decaying super-exponentially. The phases  $x^{i(\gamma_k + \gamma_j)}$  are determined by the zeta zeros, which are known numerically to arbitrary precision. The question is purely one of *cancellation*: does the sum of these oscillatory terms maintain a nonzero average, or does it cancel to  $o(x)$ ?

This is a problem in harmonic analysis, not spectral theory. The spectral part is complete:  $A$  gives the poles,  $B$  and  $C$  give the residues, and the residue structure is fully determined by antisymmetry. What remains is an exponential sum problem with Lagrange coefficients — the ancient question of whether structured oscillations can conspire to produce sustained constructive interference.

## 6 Conclusion

We have constructed the full  $(A, B, C, D)$  state-space realization of  $1/\zeta_N(s)$ , completing the spectral comb framework. The main findings:

1. **Antisymmetry forces purely imaginary residues** ( $\text{Re}(r_k) = 0$ ), giving  $B$  and  $C$  a sparse alternating structure in the modal basis.
2. **Pole weights are Lagrange interpolation weights** for the squared zero heights  $\{\gamma_k^2\}$ , connecting zeta zero distribution to classical approximation theory.
3. **The pair interaction matrix is rank-1**, meaning all pairwise pole couplings are determined by the individual weights — no independent pair structure exists.
4. **Residue products are real** (not complex), so the explicit formula coefficients carry no phase. Cancellation in the twin prime sum comes entirely from the oscillatory factors  $x^{i(\gamma_k + \gamma_j)}$ .

The transfer function perspective recasts the twin prime conjecture as a signal processing question. The system  $(A, B, C, D)$  is a multi-resonator whose output  $G(1/2 + it)$  exhibits beats from closely spaced poles. As  $t \rightarrow \infty$ , more resonators come into near-coincidence (the zero spacing shrinks like  $2\pi/\log t$ ), but each resonator’s amplitude decays super-exponentially (the Lagrange weights vanish). The twin prime conjecture asks whether this orchestra — growing in size but diminishing in volume — produces an eternal hum or eventually falls silent.

The  $A$  matrix tells you where the instruments sit.  $B$  and  $C$  tell you how loudly each one plays. The score is written; the question is whether the concert ever ends.

All constructions and numerical verifications are implemented in the Kleis formal verification language [9], with LAPACK providing the linear algebra backend. The executable source is available as `transfer_function_bc.kleis` (10 verified examples).

## 6 References

- [1] E. Atik, ‘The Riemann Zeta Function as a Transfer Function: A State-Space Perspective on Hilbert-Polya,’ Kleis Research (2025).
- [2] E. Atik, ‘The Spectral Comb: Antisymmetric Operator Architecture for the Riemann Zeta Zeros,’ Kleis Research (2025).
- [3] R. E. Kalman, ‘A new approach to linear filtering and prediction problems,’ *Journal of Basic Engineering* 82 (1960), 35-45.
- [4] M. V. Berry and J. P. Keating, ‘The Riemann zeros and eigenvalue asymptotics,’ *SIAM Review* 41 (1999), 236-266.
- [5] D. A. Goldston, J. Pintz, and C. Y. Yildirim, ‘Primes in tuples I,’ *Annals of Mathematics* 170 (2009), 819-862.
- [6] G. H. Hardy and J. E. Littlewood, ‘Some problems of partitio numerorum III: on the expression of a number as a sum of primes,’ *Acta Mathematica* 44 (1923), 1-70.
- [7] A. Connes, ‘Trace formula in noncommutative geometry and the zeros of the Riemann zeta function,’ *Selecta Mathematica* 5 (1999), 29-106.
- [8] M. Nihtila, ‘Control theory and the Riemann hypothesis: a roadmap,’ arXiv:0903.1117 (2009).
- [9] Kleis: a formal verification language for mathematics and engineering. <https://kleis.io>